

FINAL PROJECT PRESENTATION

SPOTMARK

Real-time fake news & deepfake detection

TEAM MEMBERS

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PRESENTATION AGENDA

01

Problem Statement

Defining the problem & its real-world relevance

02

Literature Review

Key background research & existing solutions

03

Methodology

Technical approach & architecture

04

Implementation

System demo & functional walkthrough

05

Testing & Validation

Results, metrics & QA

06

Innovation & Impact

What makes this unique & future scope

07

Conclusion

Summary, learnings & acknowledgements

01 | PROBLEM STATEMENT

CONTEXT



Misinformation spreads up to 6× faster than factual reporting — the fake is viral before a fact-check appears. Generative AI lets anyone create convincing fake text, images, video and voice in seconds. Impact: public health risks, financial scams, election interference and communal tension.

PROBLEM



There is no easy way to verify information instantly while reading an article or watching a video. Fakes are now hard to tell apart from real content. Fact-checks arrive late, and warnings are passive.

OBJECTIVE



Build SpotMark: a browser extension and mobile app that detects fake news and deepfakes in real time. Deliver an explainable 0–100 Credibility Score with cited sources across text, images, video and voice. Success: a verdict in seconds without interrupting browsing, plus one-click report and block.

02 | LITERATURE REVIEW & BACKGROUND

Existing Work & Key Building Blocks

- **Google Fact Check** — published claim reviews from fact-checking networks, used to corroborate claims
- **GDELT** — global news event database used for cross-corroboration
- **DistilBERT (ONNX)** — lightweight transformer for fast first-pass screening
- **Hive AI / Sensity** — commercial deepfake-detection APIs; open-source local models as a zero-cost option

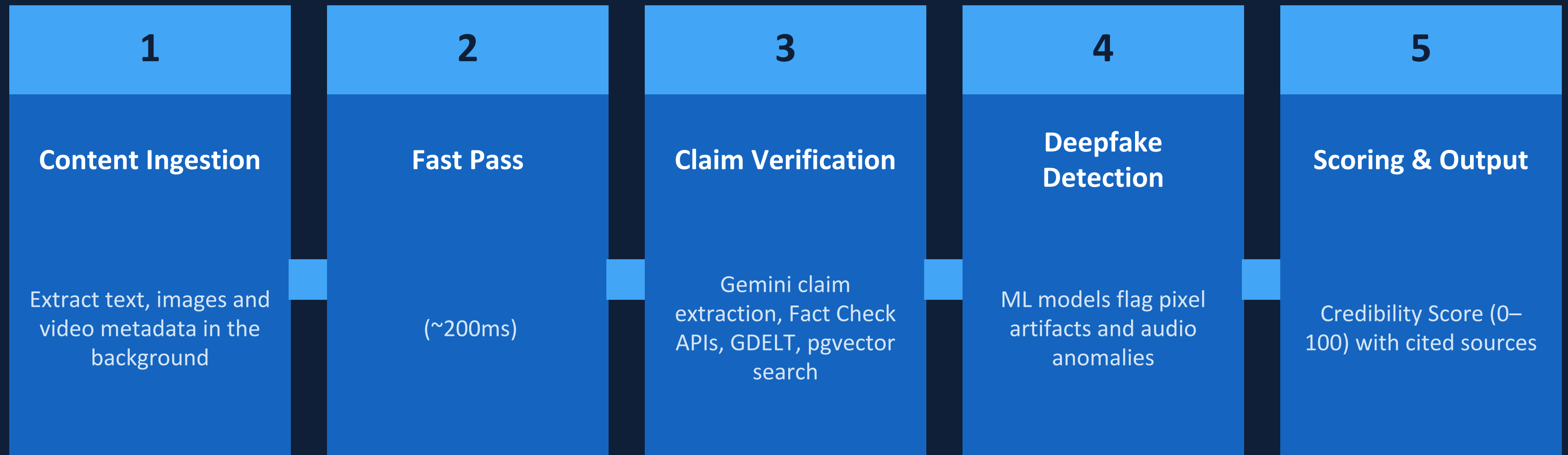
GAP IDENTIFIED

Fact-checks land after content goes viral, verdicts are often a bare True/False, and warnings are passive. Nothing offers instant, in-context checks for text and synthetic media.

OUR APPROACH

Fuse fast-pass ML, LLM claim extraction, fact-check APIs and vector search into one explainable score shown inline, with report and block actions.

03 | METHODOLOGY & TECHNICAL APPROACH



TECH STACK & TOOLS

React · TypeScript · Vite (MV3 extension) | FastAPI · Celery · pnpm + Turborepo | Gemini · DistilBERT ONNX · pgvector · GDELT · Google Fact Check | Hive AI / Sensity

03 | MONOREPO & BACKEND ARCHITECTURE

Monorepo

High-performance monorepo using pnpm workspaces and Turborepo.

extension

Chrome & Edge extension — build with React, TypeScript, Vite.

backend

FastAPI server, Celery async task queue, ML services and database adapters.

Application Entry Point

CORS- security rule
GZip compression and faster responses
X-Spotmark-Latency-MS measures how long each request takes

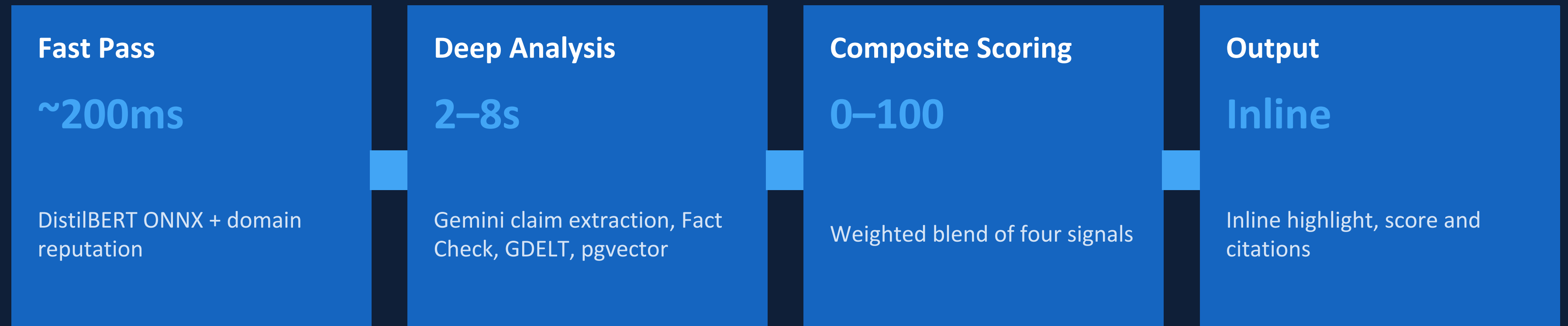
API Routes

/api/analyze-text, analyze-image, analyze-video, analyze-audio, report, block, trending, community, auth

Pluggable Deepfake Adapters

- **base_deepfake.py** — abstract interface
- **local_deepfake.py** — open-source model, zero cost
- **hive_deepfake.py** — Hive AI / Sensity, switchable via env variable

03 | ML & DETECTION PIPELINE



COMPOSITE SCORE WEIGHTS



04 | IMPLEMENTATION PROGRESS & DEMO

The screenshot shows a news article on the Aaj Tak website. The article title is "IND vs SL: Controversy in Colombo Test... Yashasvi clashes with Sri Lankan bowler, body contact also VIDEO". The text describes a heated exchange between Asitha Fernando and Yashasvi Jaiswal during a cricket test. A TruthLens overlay is positioned over the article, displaying a "47 / 100" score and a "LIKELY FALSE" status. The overlay also provides a "SCORE BREAKDOWN" and "DOMAIN REPUTATION" for the source "aajtak.in".

SCORE BREAKDOWN	
Claim Verification (50%)	50
Cross-Corroboration (15%)	0
Domain Reputation (20%)	58
Writing Style (15%)	71

DOMAIN REPUTATION	
aajtak.in	Mostly Factual
Factuality	Center
Bias Lean	Corporate
Ownership	Indian Hindi news

CLAIMS CHECKED (3)

DELIVERED SO FAR

- **Extension** — MV3, service worker, popup
- **Backend** — FastAPI scaffold & endpoints
- **Claim extraction** — Gemini, Fact Check, GDELT
- **Scoring** — Credibility engine & highlighter
- **Reputation** — Domain DB + DistilBERT
- **Moderation** — Reporting, blocking, PIB

Live on a Telugu news site: SpotMark panel shows 46/100 — Likely False, with score breakdown and domain reputation

04 | USER MODERATION & PIB ESCALATION

User Reporting API

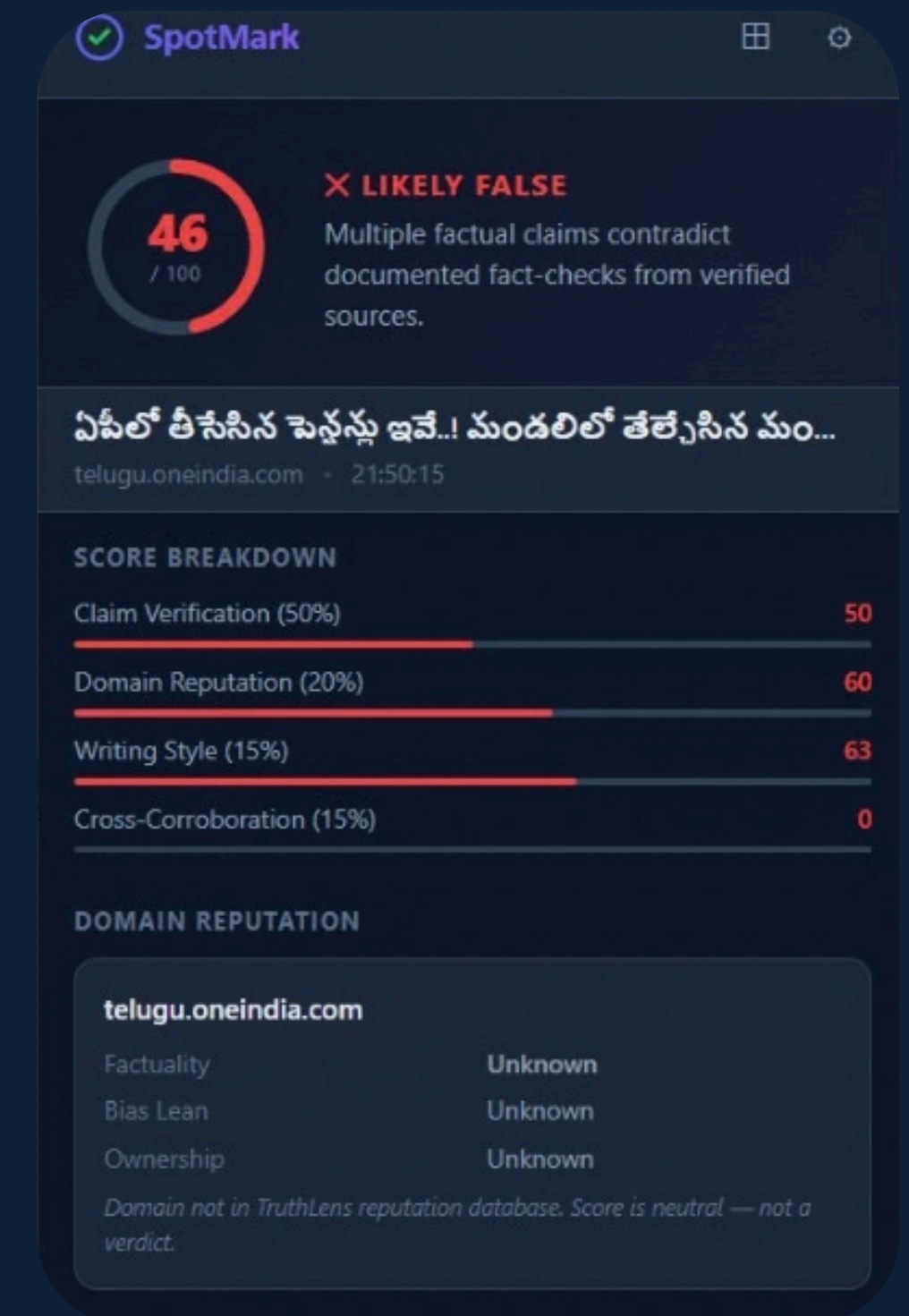
Users submit misinformation reports with category tagging — false information, clickbait, deepfake.

Official PIB Escalation

Auto-formats reports for India's PIB Fact Check: WhatsApp (+91 8799711259), email (pibfactcheck@gmail.com), portal (factcheck.pib.gov.in).

Domain & Content Blocking

Add domains or patterns to a personal blocklist; syncs across extension instances with active shield overlays.



04 | REAL-WORLD USE CASES



The WhatsApp Forward

A user receives a suspicious link or message. SpotMark flags it as a known scam or debunked claim before the user shares it.



The Viral “Leaked” Video

A shocking video of a world leader appears on social media. SpotMark instantly analyzes the media and overlays a “Deepfake Alert.”



The Student Researcher

While gathering sources, a user sees credibility scores injected directly into their browser, helping them filter out unreliable information.

05 | TESTING & VALIDATION

~200ms

Fast-pass latency

DistilBERT ONNX + domain reputation

2–8s

Deep analysis

Claim extraction, fact-check, GDELT, pgvector

0–100

Credibility Score

Four weighted signals

Testing Type	Description / Coverage	Result
Live page: Telugu news	Extension analysed a Telugu news article on a live site	46/100 · Likely False
Live page: Hindi news	Analysed an aajtak.in article with domain-reputation card	47/100 · Likely False
Backend offline	Backend unreachable: clearly shown as placeholder, not a verdict	Unverified · 50/100
Latency tracking	X-SpotMark-Latency-Ms header on every API response	Implemented

06 | INNOVATION, CREATIVITY & FUTURE SCOPE

WHAT MAKES THIS UNIQUE

- **Explainable** — shows why content is flagged, with cited sources
- **Multi-format** — text, deepfake video, cloned voice, synthetic images
- **Two-stage speed** — ~200ms fast pass, 2–8s deep analysis
- **Pluggable adapters** — local zero-cost model or Hive AI / Sensity
- **Action-oriented** — report, block or escalate to PIB

FUTURE SCOPE & ENHANCEMENTS

- React Native (Expo) mobile app & Share Sheet
- AI-generated image detection & reverse image search
- Deepfake video frame analyzer & audio voice-clone detector
- Supabase auth & cross-device sync
- Community notes, bias compass, trending feeds
- Final UI/UX polish, production packaging & demo

Trump reclassifies state-licensed medical marijuana as a less-dangerous drug in a historic shift

Left

Center

Right

 Bias Comparison

Coverage Details

Total News Sources	465
Leaning Left	112
Leaning Right	44
Center	177
Last Updated	4 days ago
Bias Distribution	53% Center

- On Thursday, Acting Attorney General Matthew Whitaker announced that the federal government will no longer prosecute state-licensed medical marijuana operators.
- President Donald Trump stated the action was a historic shift in the federal government's approach to medical treatment of patients.
- The action legitimizes standards dating to 1982 that regulated under Schedule I drugs.
- Although the order was signed in Washington, 302 of the patients.
- More than 20 Republican members of the Justice Department have signed a letter opposing the marijuana rescheduling.

 Insights by Ground AI

2 Podcasts & Opinion

 Podcast Mention

Coverage Details

Total News Sources 465

Leaning Left 112

Leaning Right 44

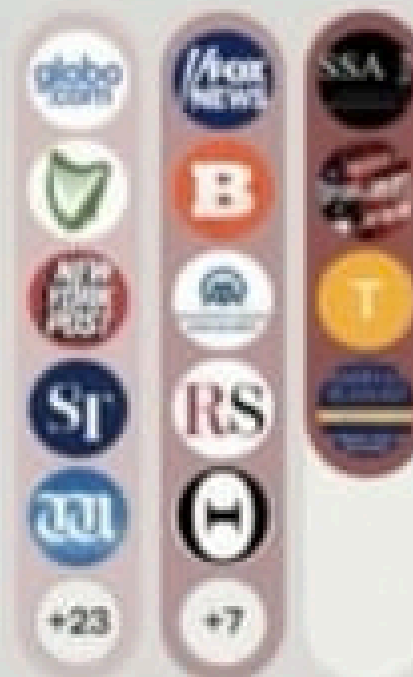
Center 177

Last Updated 4 days ago

Bias Distribution 53% Center

C 53%

L 13%



U E +123

07 | CONCLUSION & LEARNINGS

PROJECT SUMMARY

SpotMark is a real-time fake news and deepfake detector delivered as a Chrome & Edge extension, with a mobile app next. A two-stage pipeline blends LLM claim checks, fact-check APIs, GDELT, vector search and deepfake ML into an explainable 0–100 Credibility Score with cited sources.

KEY LEARNINGS

- Explainability builds trust
- Fast pass + deep analysis balances speed and depth
- Pluggable adapters limit cost and lock-in

KEY OUTCOMES

- Working MV3 extension and FastAPI backend
- Composite credibility scoring with inline highlights
- User reporting, blocking and PIB escalation

Thank You!

Questions & Feedback Welcome